

# STT 5811 Lab02: Explanatory Data Analysis

## SOLUTIONS

Monday, February 16, 2026 @ 11:56 PM

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## Instructions

This data lab is adapted from the **Intro to R - Flight Delays** lab in **Section 6.3** of the IMS2E textbook. Read the textbook lab (skip the video), then complete your analysis here. For all plots, add a title and descriptive axis labels using `labs()`. You may make other useful customizations to your plots if you would like. Use complete sentences when answering questions.

## Packages Used

Load the `tidyverse` and `openintro` packages.

```
library(tidyverse)
library(openintro)
```

# nycflights Dataset

## Dataset Description

The dataset is a random sample of flights that departed from one of three big New York City (NYC) area airports (JFK, LGA or EWR) in 2013. It was taken from a larger dataset of all 336,776 flights that departed in 2013, found in the `nycflights13` package. Each observation is one flight.

## Explore the Dataset

Assign the `nycflights` dataset to an object in your Environment.

```
flightdata <- nycflights
```

Use `names()` to view the variable names.

```
names(flightdata)
```

```
[1] "year"      "month"     "day"       "dep_time"  "dep_delay" "arr_time"
[7] "arr_delay" "carrier"   "tailnum"   "flight"    "origin"    "dest"
[13] "air_time"  "distance"  "hour"      "minute"
```

Variable definitions are important for understanding data. View the dataset's codebook by entering `?nycflights` in the Console, or search in the Help tab of the Files pane using the dataset's name. Briefly define each variable in the dataset, including the units of measure for all numerical variables. Search online for the airport names that correspond to the U.S. airport codes JFK, LGA and EWR. Your definitions do not need to be complete sentences.

- `year`: the year of the flight departure (2013 in all cases)
- `month`: the month of the flight departure
- `day`: the day of the flight departure
- `dep_time`: the time the flight departed, per local time zone
- `dep_delay`: minutes the flight's departure was delayed past its schedule departure time
- `arr_time`: the time the flight arrived, per local time zone
- `arr_delay`: minutes the flight's arrival was delayed past its schedule arrival time
- `carrier`: the two-letter abbreviation for the airline running the flight
- `tailnum`: the tail number of the plane for the flight
- `flight`: the flight number for the flight
- `origin`: the airport from which the flight departed
- `dest`: the airport to which the flight was headed
- `air_time`: minutes the plane spend in the air for that flight
- `distance`: miles the plane flew on that flight
- `hour`: the hour of the flight departure time (combines with `minute` to get the full time)
- `minute`: the minute of the flight departure time (combines with `hour` to get the full time)
- JFK: John F. Kennedy International Airport
- LGA: LaGuardia Airport
- EWR: Newark Liberty International Airport

Note: Other wordings are possible/acceptable for these definitions.

Create a new variable that identifies departures as “on-time” or “delayed,” where any flight delayed less than five minutes is considered “on-time.” Create a similar variable for arrivals, where “on time” is any flight that was less than five minutes late. Also make a variable that represents the average speed traveled by the plane for each flight, in miles per hour (mph).

```
flightdata <- flightdata |>
  mutate(
    dep_type = ifelse(dep_delay < 5, "on-time", "delayed"),
    arr_type = ifelse(arr_delay < 5, "on-time", "delayed"),
    avg_speed = distance/(air_time/60)
  )
```

Use `glimpse()` to explore the dataset and verify your new variables.

```
glimpse(flightdata)

Rows: 32,735
Columns: 19
$ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, ~
$ month     <int> 6, 5, 12, 5, 7, 1, 12, 8, 9, 4, 6, 11, 4, 3, 10, 1, 2, 8, 10~
$ day       <int> 30, 7, 8, 14, 21, 1, 9, 13, 26, 30, 17, 22, 26, 25, 21, 23, ~
$ dep_time  <int> 940, 1657, 859, 1841, 1102, 1817, 1259, 1920, 725, 1323, 940~
$ dep_delay <dbl> 15, -3, -1, -4, -3, -3, 14, 85, -10, 62, 5, 5, -2, 115, -4, ~
$ arr_time  <int> 1216, 2104, 1238, 2122, 1230, 2008, 1617, 2032, 1027, 1549, ~
$ arr_delay <dbl> -4, 10, 11, -34, -8, 3, 22, 71, -8, 60, -4, -2, 22, 91, -6, ~
$ carrier   <chr> "VX", "DL", "DL", "DL", "9E", "AA", "WN", "B6", "AA", "EV", ~
$ tailnum   <chr> "N626VA", "N3760C", "N712TW", "N914DL", "N823AY", "N3AXAA", ~
$ flight    <int> 407, 329, 422, 2391, 3652, 353, 1428, 1407, 2279, 4162, 20, ~
$ origin    <chr> "JFK", "JFK", "JFK", "JFK", "LGA", "LGA", "EWR", "JFK", "LGA~
$ dest      <chr> "LAX", "SJU", "LAX", "TPA", "ORF", "ORD", "HOU", "IAD", "MIA~
$ air_time  <dbl> 313, 216, 376, 135, 50, 138, 240, 48, 148, 110, 50, 161, 87,~
$ distance  <dbl> 2475, 1598, 2475, 1005, 296, 733, 1411, 228, 1096, 820, 264,~
$ hour      <dbl> 9, 16, 8, 18, 11, 18, 12, 19, 7, 13, 9, 13, 8, 20, 12, 20, 6~
$ minute    <dbl> 40, 57, 59, 41, 2, 17, 59, 20, 25, 23, 40, 20, 9, 54, 17, 24~
$ dep_type  <chr> "delayed", "on-time", "on-time", "on-time", "on-time", "on-t~
$ arr_type  <chr> "on-time", "delayed", "delayed", "on-time", "on-time", "on-t~
$ avg_speed <dbl> 474.4409, 443.8889, 394.9468, 446.6667, 355.2000, 318.6957, ~
```

## Categorical Analysis

### Departure Airports

Create a table to show how many flights departed from each airport. Include counts and proportions.

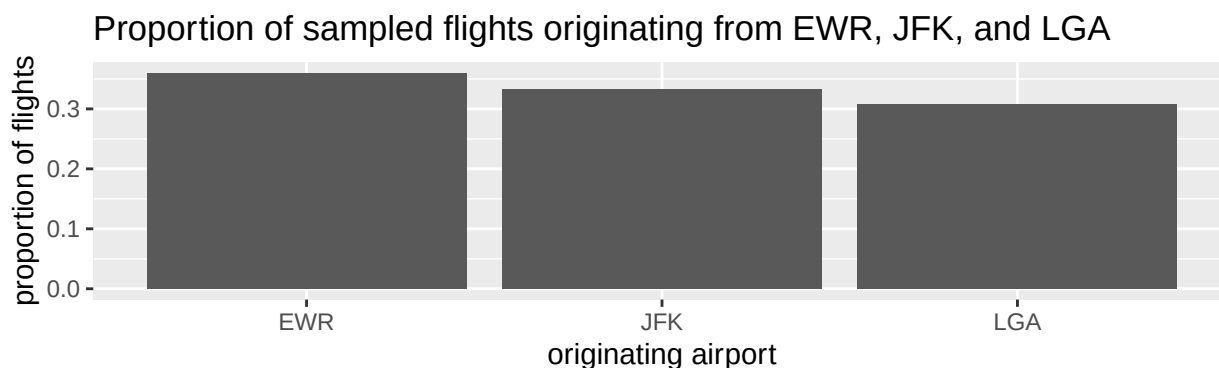
```
flightdata |>
  count(origin) |>
  mutate(prop = prop.table(n))
```

```
# A tibble: 3 x 3
```

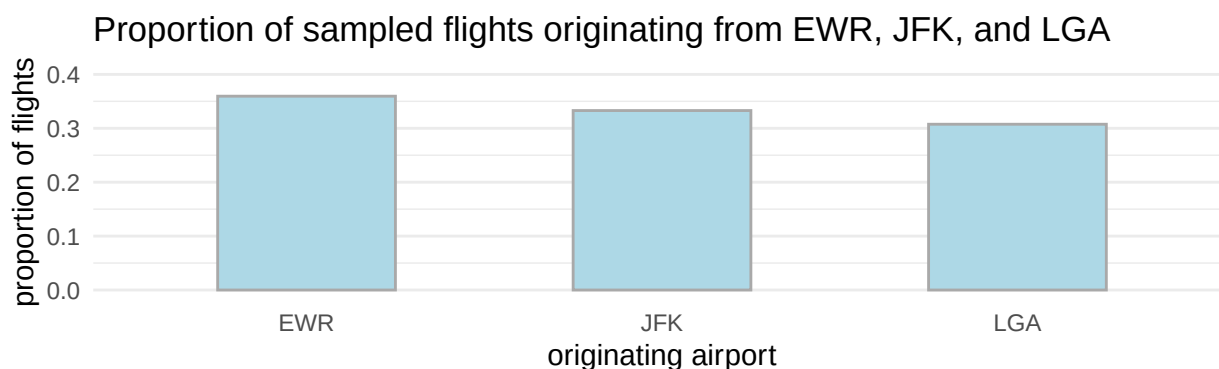
|   | origin | n     | prop  |
|---|--------|-------|-------|
|   | <chr>  | <int> | <dbl> |
| 1 | EWB    | 11771 | 0.360 |
| 2 | JFK    | 10897 | 0.333 |
| 3 | LGA    | 10067 | 0.308 |

Use the `geom_bar()` function to make a barplot showing the proportion of flights from each airport.

```
# Minimal required
ggplot(flightdata, aes(x = origin, y = after_stat(prop), group = 1)) +
  geom_bar() +
  labs(
    title = "Proportion of sampled flights originating from EWR, JFK, and LGA",
    x = "originating airport",
    y = "proportion of flights"
  )
```



```
# Added formatting
ggplot(flightdata, aes(x = origin, y = after_stat(prop), group = 1)) +
  geom_bar(width = 0.5, color = "darkgray", fill = "lightblue") +
  scale_y_continuous(limits = c(0, 0.4)) +
  theme_minimal() +
  theme(panel.grid.major.x = element_blank()) +
  labs(
    title = "Proportion of sampled flights originating from EWR, JFK, and LGA",
    x = "originating airport",
    y = "proportion of flights"
  )
```



## Destination Airports

Use the `n_distinct()` function to find how many different destination airports the data contain.

```
n_distinct(flightdata$dest)
```

```
[1] 102
```

```
flightdata |> pull(dest) |> n_distinct()
```

```
[1] 102
```

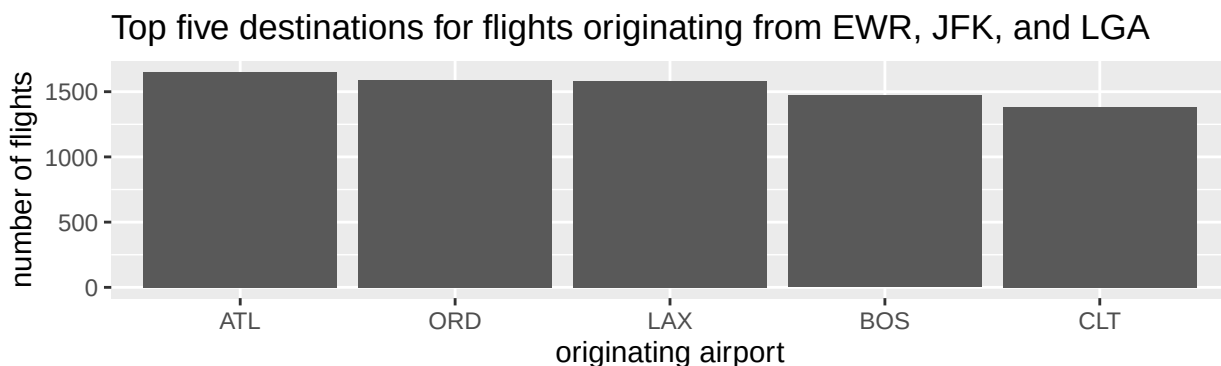
Make a table to show the top five destinations for departing flights. Include counts and proportions. Save the result as an object in your Environment, but be sure to display it here as well.

```
top5dest <- flightdata |>
  count(dest) |>
  mutate(prop = prop.table(n)) |>
  slice_max(n, n = 5)
top5dest
```

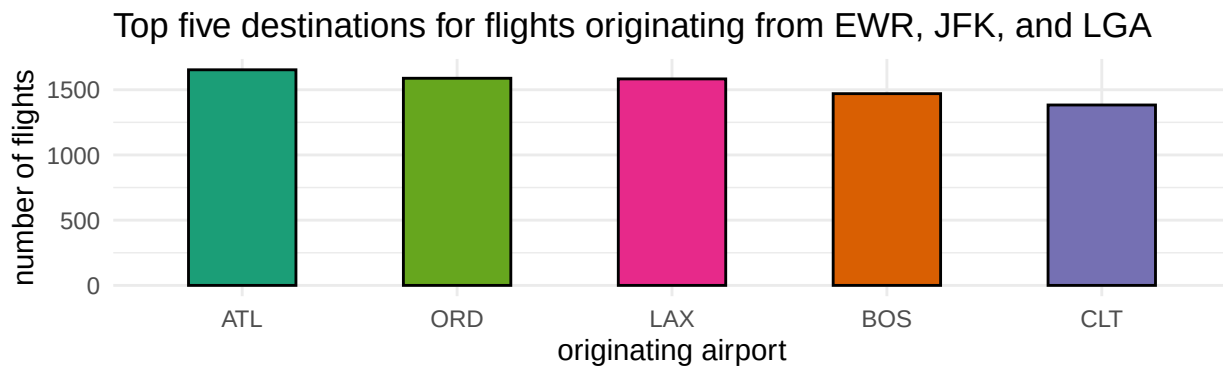
```
# A tibble: 5 x 3
  dest      n  prop
  <chr> <int> <dbl>
1 ATL    1653 0.0505
2 ORD    1588 0.0485
3 LAX    1583 0.0484
4 BOS    1470 0.0449
5 CLT    1383 0.0422
```

Use your saved table and the `geom_col()` function to create a barplot for the top five destinations. Use `reorder()` in the plot aesthetics to sort the bars from largest to smallest.

```
# Minimal required
ggplot(top5dest, aes(x = reorder(dest, -n), y = n)) +
  geom_col() +
  labs(
    title = "Top five destinations for flights originating from EWR, JFK, and LGA",
    x = "originating airport",
    y = "number of flights"
  )
```



```
# Added formatting
ggplot(top5dest, aes(x = reorder(dest, -n), y = n)) +
  geom_col(width = 0.5, color = "black", aes(fill = dest)) +
  theme_minimal() +
  theme(legend.position = "none") +
  scale_fill_brewer(palette = "Dark2") +
  labs(
    title = "Top five destinations for flights originating from EWR, JFK, and LGA",
    x = "originating airport",
    y = "number of flights"
  )
```



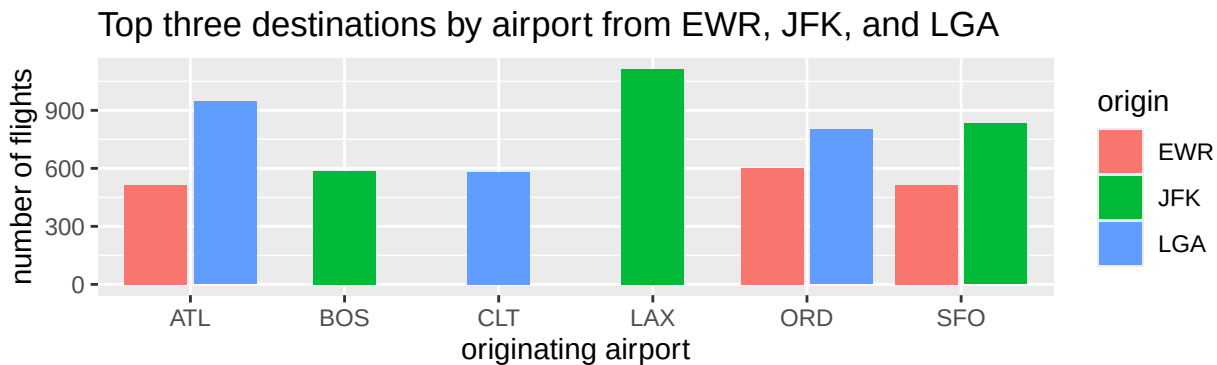
Make a table to show the top three destinations grouped by airport. Include counts and proportions. Save the result as an object in your Environment, but be sure to display it here as well.

```
top3dest <- flightdata |>
  group_by(origin) |>
  count(dest) |>
  mutate(prop = prop.table(n)) |>
  slice_max(n, n = 3)
top3dest
```

```
# A tibble: 9 x 4
# Groups:   origin [3]
  origin dest      n  prop
  <chr>  <chr> <int> <dbl>
1 EWR    ORD     601 0.0511
2 EWR    SFO     513 0.0436
3 EWR    ATL     512 0.0435
4 JFK    LAX    1115 0.102
5 JFK    SFO     832 0.0764
6 JFK    BOS     585 0.0537
7 LGA    ATL     946 0.0940
8 LGA    ORD     801 0.0796
9 LGA    CLT     583 0.0579
```

Using `geom_col(position = position_dodge2(preserve = "single"))`, and `fill = origin` in the overall plot aesthetics, create a barplot for the top three destinations grouped by airport.

```
# Minimal required
ggplot(top3dest, aes(x = dest, y = n, fill = origin)) +
  geom_col(position = position_dodge2(preserve = "single")) +
  labs(
    title = "Top three destinations by airport from EWR, JFK, and LGA",
    x = "originating airport",
    y = "number of flights"
  )
```



## Analysis Discussion 1

1. How do total flight departures compare across the three airports? Discuss this as if I cannot see your table or plot (e.g., cite relevant summary statistics)

About one-third of the flights originate from each airport. More specifically, a little over 35% take off from EWR, a little under 35% take off from JFK, and about 30% take off from LGA.

2. What are the top five flight destinations? What fraction of flights do they represent overall? Does this set of destinations make sense to you? Discuss.

The top five destinations are (1 to 5): Atlanta (ATL), New Orleans (ORD), Los Angeles (LAX), Boston (BOS) and Charlotte (CLT). They represent about 23% of all flights from EWR, JFK, and LGA. This set of destinations makes sense. They are all in major cities, and serve as transfer hubs and/or international departure points.

3. Discuss similarities and differences between the top three destinations for each airport, and make relevant comparisons to the top five overall airports.

When looking at the top three destinations for each airport, the set includes all of the overall top five destinations found above, plus San Francisco (SFO). However, each airport has a different set. BOS, CLT, and LAX are among the top three destinations for only one airport each, while ATL, ORD, and SFO are each on two airports' lists.

- EWR's top three are ORD, ATL, and SFO, with roughly equal numbers of flights (~500 to 600 each).
- JFK's ordered top three are LAX, SFO, and BOS. LAX has about twice as many flights (~1200) as BOS (~600), with SFO halfway between.
- LGA's ordered top three are ATL, ORD, and CLT. CLT has about two-thirds as many flights (~600) as ATL (~900), with ORD halfway between.

# Numerical Analysis

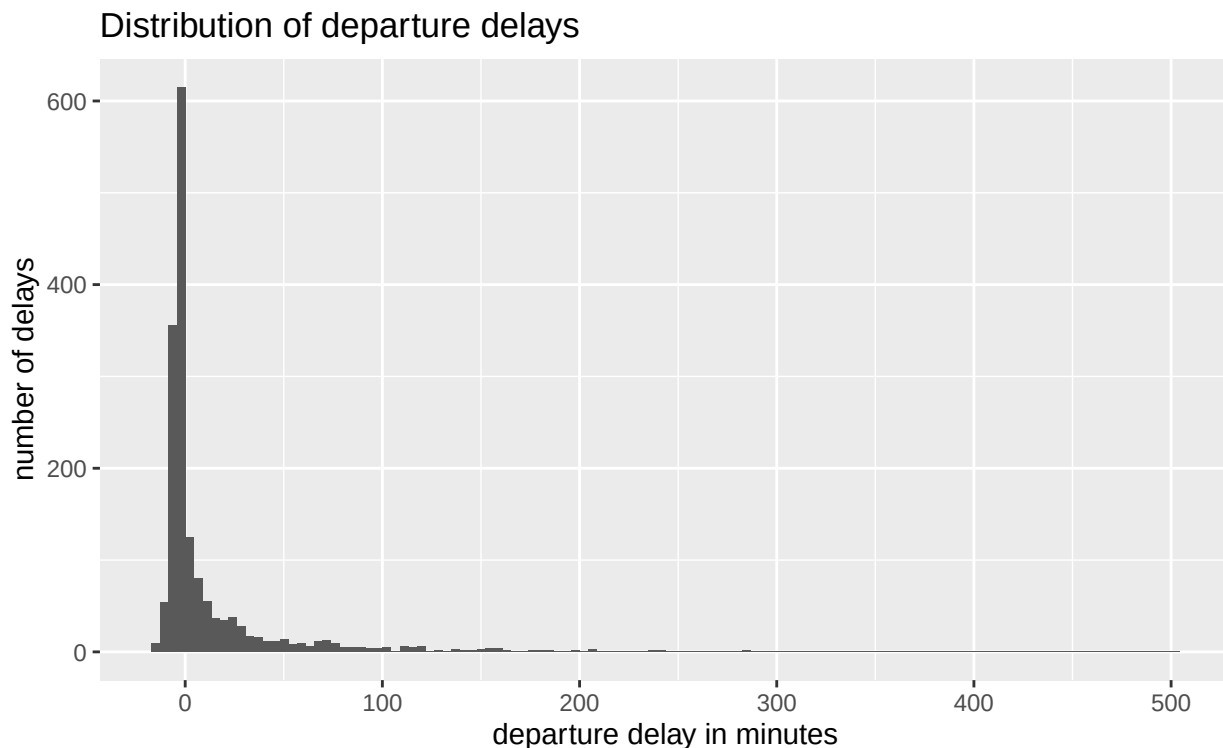
## Departure Delays

Filter the dataset to extract only those flights going to the most frequent destination (overall). Use this extracted set of flights for all analyses involving delays.

```
flightdataATL <- flightdata |>
  filter(dest == "ATL")
```

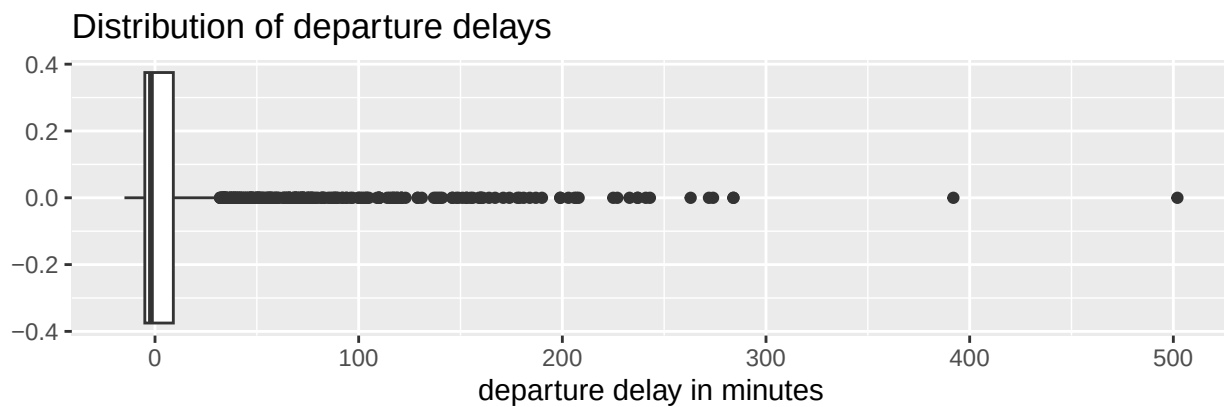
Make a histogram and boxplot to visualize the distribution of departure delays and identify outliers. Use 120 bins for your histogram.

```
# Minimal required
ggplot(flightdataATL, aes(x = dep_delay)) +
  geom_histogram(bins = 120) +
  labs(
    title = "Distribution of departure delays",
    x = "departure delay in minutes",
    y = "number of delays"
  )
```



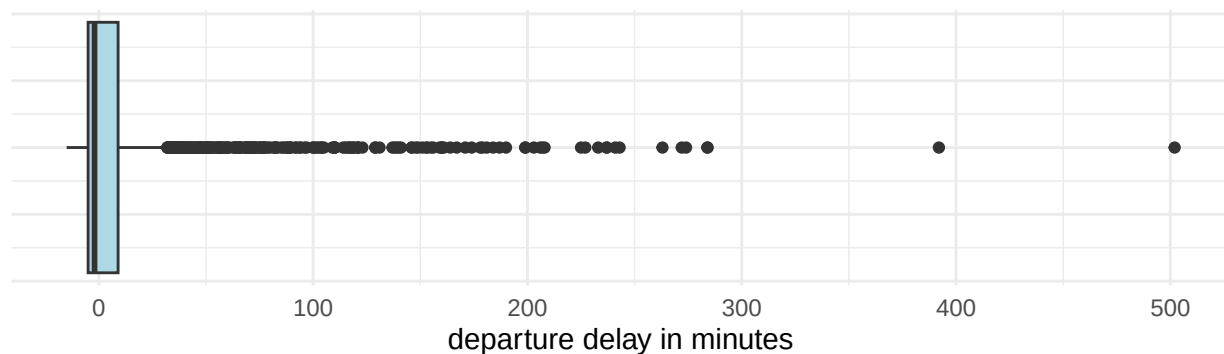
```
# Minimal required
ggplot(flightdataATL, aes(x = dep_delay)) +
  geom_boxplot() +
  labs(
    title = "Distribution of departure delays",
    x = "departure delay in minutes"
  )
```





```
# Added formatting
ggplot(flightdataATL, aes(x = dep_delay)) +
  geom_boxplot(fill = "lightblue") +
  labs(
    title = "Distribution of departure delays",
    x = "departure delay in minutes"
  ) +
  theme_minimal() +
  theme(
    axis.title.y = element_blank(),
    axis.text.y = element_blank(),
  )
```

### Distribution of departure delays



Calculate the mean and median departure delay, as well as the standard deviation, IQR, and upper and lower fences used to identify outliers in the boxplot.

```
flightdataATL |>
  summarize(
    mean_dd = mean(dep_delay),
    med_dd = median(dep_delay),
    sd_dd = sd(dep_delay),
    iqr_dd = IQR(dep_delay),
    lf_dd = fivenum(dep_delay)[2] - iqr_dd,
    uf_dd = fivenum(dep_delay)[4] + iqr_dd
  )
```

```
# A tibble: 1 x 6
  mean_dd med_dd sd_dd iqr_dd lf_dd uf_dd
  <dbl>   <dbl> <dbl>   <dbl> <dbl> <dbl>
1   13.0     -2  41.8    14   -19    23
```

Calculate the same statistics, along with the number of flights, grouped by the origin airport.

```
flightdataATL |>
  group_by(origin) %>%
  summarize(
    n_origin = n(),
    mean_dd = mean(dep_delay),
    med_dd = median(dep_delay),
    sd_dd = sd(dep_delay),
    iqr_dd = IQR(dep_delay),
    lf_dd = fivenum(dep_delay)[2] - iqr_dd,
    uf_dd = fivenum(dep_delay)[4] + iqr_dd
  )
```

```
# A tibble: 3 x 8
  origin n_origin mean_dd med_dd sd_dd iqr_dd lf_dd uf_dd
  <chr>   <int>   <dbl>   <dbl> <dbl>   <dbl> <dbl> <dbl>
1 EWR       512    16.6     -1  47.2    21   -25    38
2 JFK       195     8.59     -1  27.1    10   -13    17
3 LGA       946    11.9     -2  41.1    11   -16    17
```

Find the on-time departure rates for the three origin airports. Arrange them from best to worst.

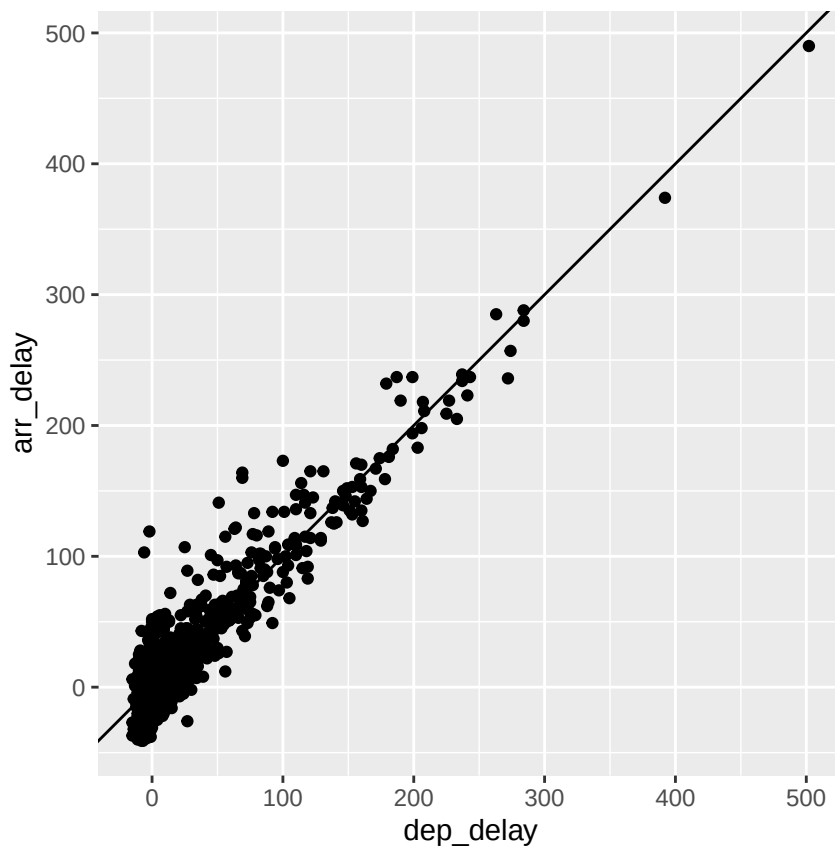
```
flightdataATL |>
  group_by(origin) |>
  summarise(ot_dep_rate = sum(dep_type == "on-time") / n()) |>
  arrange(ot_dep_rate)
```

```
# A tibble: 3 x 2
  origin ot_dep_rate
  <chr>   <dbl>
1 EWR       0.662
2 JFK       0.703
3 LGA       0.722
```

## Departure vs. Arrival Delays

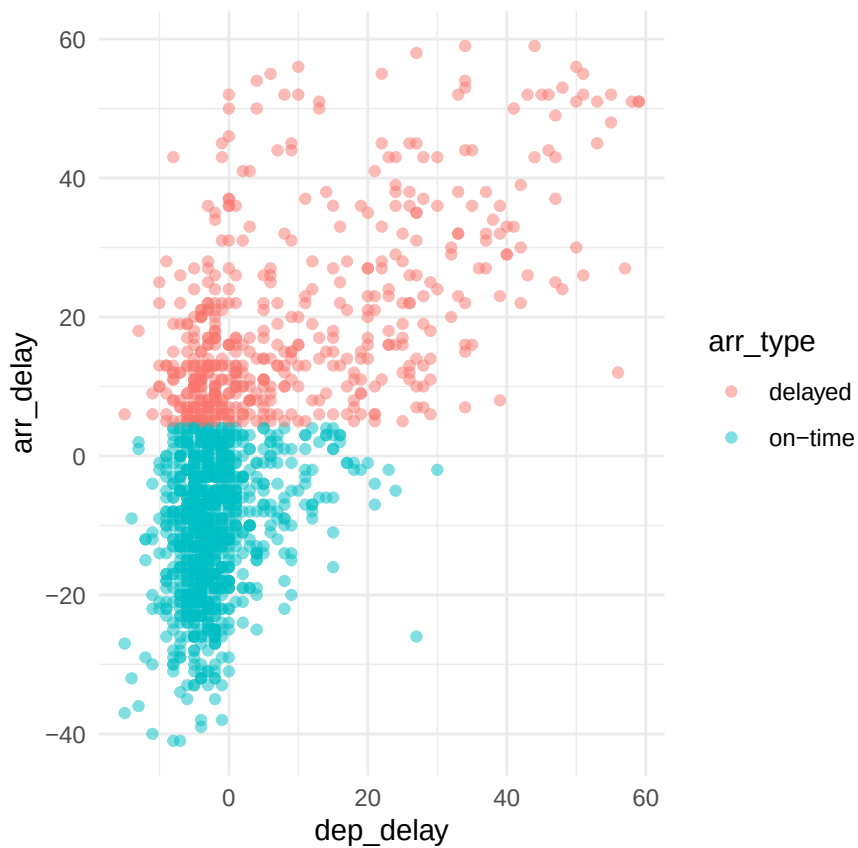
Plot arrival delays (y) versus departure delays (x). Add a line showing  $y = x$  using `geom_abline()`. Use `coord_fixed()` so that the x- and y-axis units have the same scaling.

```
flightdataATL |>
  ggplot(aes(x = dep_delay, y = arr_delay)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0) +
  coord_fixed()
```



Zoom in on the lower corner of the plot by filtering for cases where both the departure and arrival delays are less than one hour, or by adjusting the axes to only show those observations. Color the points by whether or not the arrival was delayed. Use `alpha` to better visualize overlapping points.

```
flightdataATL |>
  filter(dep_delay < 60, arr_delay < 60) |>
  ggplot(aes(x = dep_delay, y = arr_delay, color = arr_type)) +
  geom_point(alpha = 0.5) +
  coord_fixed() +
  theme_minimal()
```

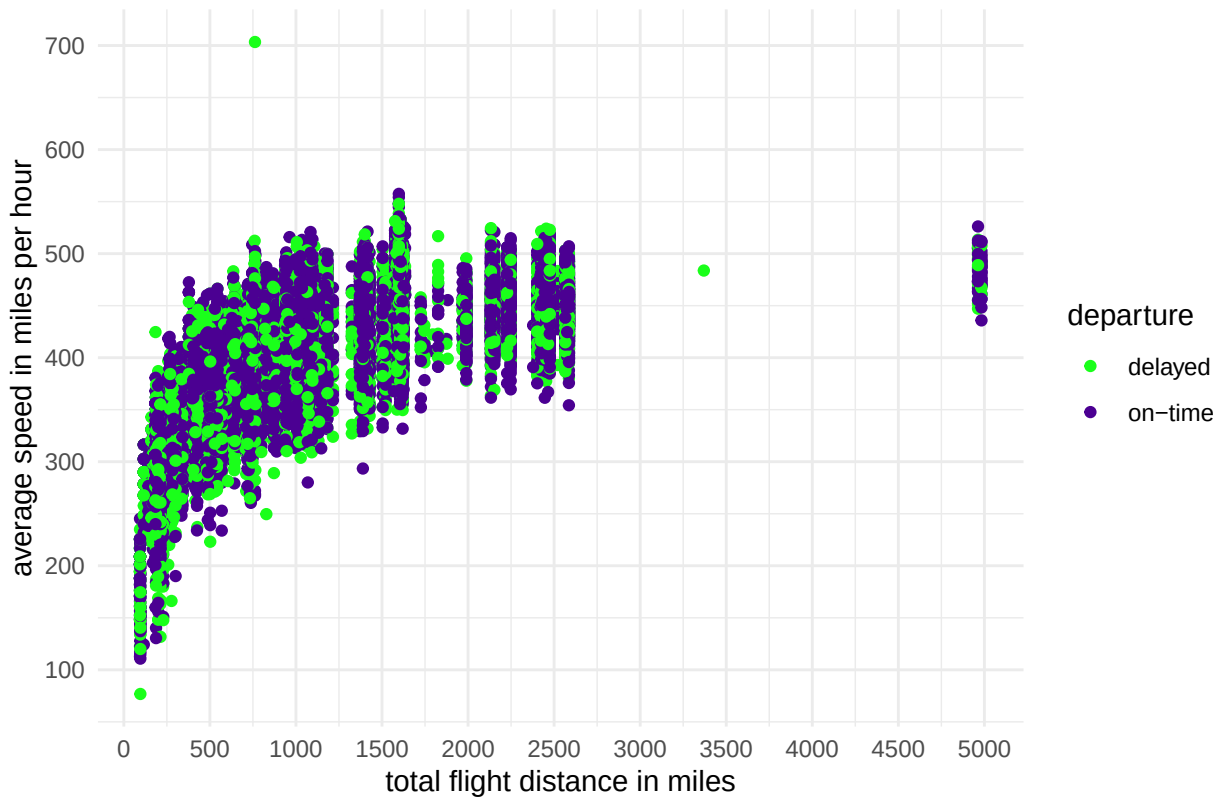


## Flight Speed vs. Distance

Plot average speed (y) versus distance (x) for the full dataset. Color the points by whether departure was on-time or delayed.

```
ggplot(flightdata, aes(x = distance, y = avg_speed, color = dep_type)) +
  geom_point() +
  scale_color_manual(values = c("#1AFF1A", "#4B0092")) +
  scale_x_continuous(breaks = seq(0, 5000, 500)) +
  scale_y_continuous(breaks = seq(0, 700, 100)) +
  theme_minimal() +
  labs(
    title = "average speed of flights departing from EWR, JFK, and LGA in 2013",
    x = "total flight distance in miles",
    y = "average speed in miles per hour",
    color = "departure"
  )
```

average speed of flights departing from EWR, JFK, and LGA in 2013



## Analysis Discussion 2

1. There are flights in this dataset with negative departure delay times. What does that mean?

ERASE AND TYPE YOUR ANSWER HERE

2. What is the distribution shape for departure delay times? Why does this shape make sense?

ERASE AND TYPE YOUR ANSWER HERE

3. What do the mean and median each tell us about the length of a “typical” flight delay to the chosen destination? Which is a better measure of center, in context? Explain your reasoning.

ERASE AND TYPE YOUR ANSWER HERE

4. If we consider only on-time departure rates and variability in departure delays to the chosen destination, which airport seems to be the best one to fly out of? Explain your reasoning.

ERASE AND TYPE YOUR ANSWER HERE

5. Is there a relationship between departure delays and arrival delays to the chosen destination? If so, describe it. What is the cutoff point for departure delays where you can still reasonably expect to get to the destination on time? Explain your reasoning.

ERASE AND TYPE YOUR ANSWER HERE

6. Describe the relationship between average speed and distance. Given the context, why does this shape make sense? Does average speed seem to be influenced by whether or not a flight was delayed? Explain your reasoning.

# Session Information

```
sessionInfo()
```

R version 4.5.2 (2025-10-31)

Platform: x86\_64-redhat-linux-gnu

Running under: Red Hat Enterprise Linux 9.7 (Plow)

Matrix products: default

BLAS/LAPACK: FlexiBLAS OPENBLAS-OPENMP; LAPACK version 3.9.0

locale:

```
[1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
[3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
[5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
[7] LC_PAPER=en_US.UTF-8     LC_NAME=C
[9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

time zone: America/New\_York

tzcode source: system (glibc)

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] openintro_2.5.0    usdata_0.3.1      cherryblossom_0.1.0
[4] airports_0.1.0     lubridate_1.9.4    forcats_1.0.1
[7] stringr_1.5.2      dplyr_1.1.4       purrr_1.1.0
[10] readr_2.1.5        tidyr_1.3.1       tibble_3.3.0
[13] ggplot2_4.0.0      tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] gtable_0.3.6      compiler_4.5.2    tinytex_0.57      tidyselect_1.2.1
[5] dichromat_2.0-0.1 scales_1.4.0      yaml_2.3.10       fastmap_1.2.0
[9] R6_2.6.1          labeling_0.4.3    generics_0.1.4    knitr_1.50
[13] pillar_1.11.1     RColorBrewer_1.1-3 tzdb_0.5.0        rlang_1.1.6
[17] utf8_1.2.6        stringi_1.8.7     xfun_0.53         S7_0.2.0
[21] timechange_0.3.0   cli_3.6.5         withr_3.0.2       magrittr_2.0.4
[25] digest_0.6.37     grid_4.5.2        rstudioapi_0.17.1 hms_1.1.4
[29] lifecycle_1.0.4    vctrs_0.6.5       evaluate_1.0.5    glue_1.8.0
[33] farver_2.1.2      rmarkdown_2.30    tools_4.5.2       pkgconfig_2.0.3
[37] htmltools_0.5.8.1
```